



Google now owns SCHAFT

The best bot built at the DARPA Robotics Challenge is now to be funded by Google, its new owner <http://dgit.in/SCHAFT>



Vicarious raises \$40 Mn

Secretive artificial intelligence firm gets funding from the likes of Mark Zuckerberg, and Elon Musk <http://dgit.in/Vic40Mil>

"The robots we're all worried about may be us!"

Asst. Professor James McLurkin from Rice University is a lifelong robotics fan. We had a great conversation with him last month on his maiden trip to India on all things robotics related. What's more, we also tickled his fancy and got his thoughts on the future of voice-based assistants, machine learning, the possibility of a humanoid butler walking around in our homes, and the moral implications of creating sentient AI and robots. Excerpts from our exclusive interview:

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What sparked your interest in robotics?

JM: For me, robotics is the culmination of a lifetime spent building better toys. When I was younger, cardboard boxes, model trains, and as I got older radio-controlled cars, video games, and lots of LEGO. Ultimately, if you put that much technology into a 15-16 year old, you move towards building robots – that was the most natural inclination for me. I still like to think myself as building better toys that everyone can use. So, robots for everybody, yes.

What sort of robots do you build and study?

JM: Right now, we have four different robots split into two different types. This particular robot was built at Rice University – in collaboration with our students – and the goal of this robot is to have research grade features and qualities but built as cheap as possible to minimize costs. The ultimate goal of my work is to understand distributed algorithms for multi-robot systems. So if somebody gave me 10,000 robots, I need to figure out how to put them together, how to program them, how to get them to do tasks as a group.

And we see examples of this all over. Ants, bees, termites, wasps, all of these insect communities do very complicated tasks as groups, that individuals can't do by themselves. We don't want to copy nature, because nature's far too sophisticated for us to copy, but understand the algorithmic implications of these tasks – what's the



James McLurkin, Asst. Professor, Dept. of Computer Science, Rice University

input, what's the software, and what's the output of these tasks that nature runs, so to speak. There are various things that insects do in their lifetime, which are useful to us if we get robots to do them, too.

How will your technology, research and robots be implemented in the real-world?

JM: As robots become universal, more practical, cheap and accessible, it will be uncommon to see individual robots working on a task and groups of robots will be common. I want to take that concept to the extreme where I'm not worried about 10-20 robots but 10,000-20,000 robots. Enough robots to build buildings, to reclaim land, to search the oceans for the giant squid and other unexplored creatures. Right now we have two robots on

Mars, what if we had 2,000? We could definitely search the planet far more efficiently.

Do you think there's a lack of awareness of the impact of robots on our lives among the general public?

JM: We as humans have two views of robots: the classic view concerning the robot butler and then we have the giant killer robots (Terminator bots). Most of us won't see either of those possibilities for a very long time, and hopefully we'll never get to the giant killer robot stage. But we're going to see stuff like Google's robot cars, which's a hot topic right now, and this is a problem we'll see getting solved within our lifetime (of a self-driving car). But robots are all around us, in a way. I flew here in what's essentially a very large robot called a 777, my dishwasher is robotic. So that inherent need to sense, compute, act in the physical world without human intervention is one of the keys to robotics.

The other area where we're working on actively as a community is artificial intelligence (AI). In the example of Google's self-driving car, there are basic things that the car needs to understand, but it doesn't need to be very smart. It does need to be careful and good at driving, but it doesn't need to be creative, draw inferences from experiences that are new and novel. These are things that we just don't know how to do yet. The type of intelligence we will be able to achieve is one that's given a large amount of data, and analysing that data to find the best possible solution to a problem. But there's this whole other aspect of intel-